

CaptionAI: AI-Powered Social Media Caption Generator

Project Description:

Credit Card Approval Prediction is a Machine Learning-based web application developed to assist users in predicting the likelihood of credit card approval based on applicant information. The application analyzes demographic, employment, and financial details using a trained Machine Learning model and provides instant prediction results through an interactive web interface.

The system is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript**. It includes features such as applicant data validation, credit card approval prediction, prediction history management, and a Machine Learning models information page. The application has been deployed on **Render**, making it accessible through a web browser for demonstration and evaluation.

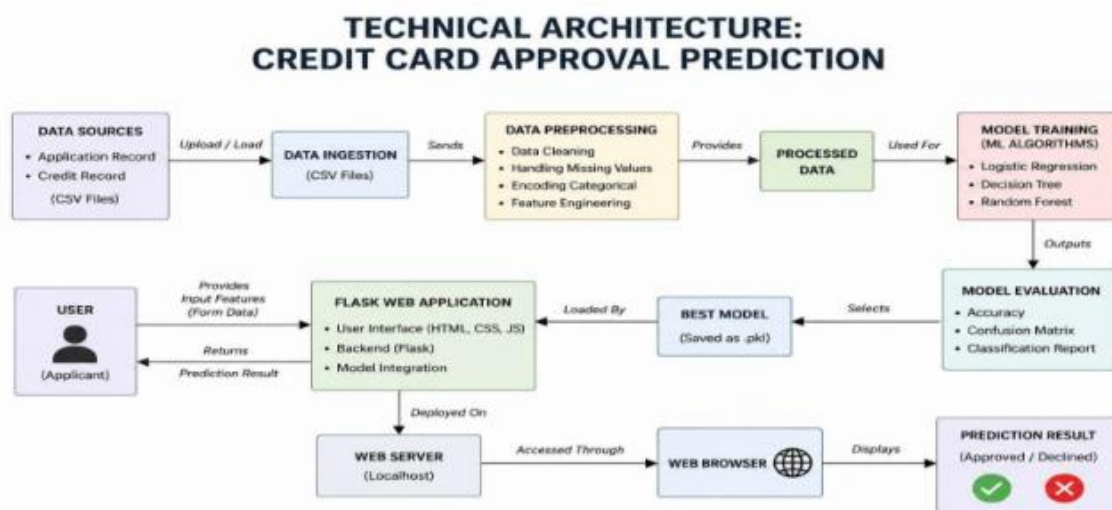
Scenarios

Scenario 1: A user enters personal, employment, and financial information into the prediction form. The application processes the input using the trained Machine Learning model and instantly displays whether the credit card application is likely to be approved or rejected.

Scenario 2: A bank employee uses the application to evaluate multiple customer applications efficiently. The system provides consistent prediction results, helping reduce manual verification time and supporting preliminary decision-making.

Scenario 3: Students and researchers use the application to understand how Machine Learning can be applied in financial decision support systems. The application demonstrates the complete workflow from data input to prediction while maintaining a history of previous predictions.

Technical Architecture:



Description:

The CredAI application follows a modular three-layer architecture.

- **Presentation Layer:** Responsive web interface developed using HTML, CSS, JavaScript, and Jinja2 templates.
- **Application Layer:** Flask backend responsible for routing, request processing, input validation, prediction management, and history handling.
- **Machine Learning Layer:** Preprocessing components and trained Machine Learning model used to generate credit card approval predictions.

This architecture improves maintainability, scalability, and efficient communication between the user interface and the prediction model.

Pre-requisites:

- Python Programming
 - Flask Framework
 - HTML, CSS, and JavaScript
 - Machine Learning Fundamentals
 - Scikit-learn
 - Pandas and NumPy
 - Git and GitHub
 - Virtual Environment Management
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Project Workflow

Milestone 1: Environment Setup and Architecture Design

Activity 1.1: Requirement Analysis

- Study project requirements.
- Identify user needs.
- Define project objectives.

Activity 1.2: Architecture Design

- Design the application architecture.
- Define interactions between frontend, backend, machine learning model, and database.

Activity 1.3: Development Environment Setup

- Install Python.
- Create a virtual environment.
- Install project dependencies.
- Configure Flask.

Activity 1.4: Project Structure

Create folders for:

- templates/
- static/

- model/
 - data/
 - notebook/
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Milestone 2: Core Functionalities Development

Activity 2.1: Flask Backend Development

Create application routes:

- /prediction
- /result
- /history
- /ml-models

Responsibilities:

- Process user requests.
- Validate input.
- Connect frontend with Machine Learning model.
- Manage navigation between pages.

Activity 2.2: Data Processing Module

Responsibilities:

- Validate applicant information.
- Encode categorical values.
- Scale required numerical features.
- Prepare model input.

Activity 2.3: Prediction Module

Responsibilities:

- Load trained Machine Learning model.
- Generate approval prediction.
- Return prediction result.

Activity 2.4: Prediction History Module

Responsibilities:

- Store prediction records.
 - Retrieve previous prediction history.
 - Display prediction history through the History page.
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Milestone 3: App.py Development

Activity 3.1: User Interface Development

Develop pages for:

- Home
- Prediction
- Result
- History
- ML Models
- Features

- How It Works

Activity 3.2: User Experience Enhancement

Implement:

- Responsive Design
 - Form Validation
 - Smooth Navigation
 - Interactive User Interface
 - Error Handling
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Milestone 4: Frontend Development

Activity 4.1: Prediction History Management

Store prediction records in the SQLite database.

Responsibilities:

- Save prediction history.
- Retrieve previous predictions.
- Display stored records.

Activity 4.2: Machine Learning Model Management

Store trained model files.

Responsibilities:

- Load trained model.
 - Load scaler and encoders.
 - Perform prediction efficiently.
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Milestone 5: Deployment

Activity 5.1: Functional Testing

Verify:

- Prediction functionality.
- User input validation.
- Navigation.
- History module.
- Machine Learning integration.

Activity 5.2: Application Testing

Run application locally:

```
python app.py
```

Verify:

- Home Page
- Prediction Page
- Result Page
- History Page
- ML Models Page

Milestone 6: Conclusion

Activity 6.1: Cloud Deployment

Deploy the application using **Render**.

Verify:

- Application accessibility.
- Prediction functionality.
- Responsive user interface.

Activity 6.2: Version Control

Upload the complete project to GitHub.

Repository includes:

- Source Code
- Documentation
- Model Files
- Dataset
- README
- Deployment Configuration

Exploring the Application Pages

Home Page

Description:

Provides an introduction to CredAI and allows users to navigate to different sections of the application.

Prediction Page

Description:

Allows users to enter applicant information and submit the details for credit card approval prediction.

Result Page

Description:

Displays the Machine Learning prediction indicating whether the application is likely to be approved or rejected.

History Page

Description:

Displays previously generated prediction records stored in the application's SQLite database.

ML Models Page

Description:

Provides information about the Machine Learning model used for prediction, including the model's purpose and performance metrics.

Features Page

Description:

Highlights the major features of the CredAI application, including prediction, history tracking, responsive design, and Machine Learning integration.

How It Works Page

Description:

Explains the complete workflow of the application, from user input and data preprocessing to Machine Learning prediction and result generation.

Conclusion

CredAI demonstrates the practical use of Machine Learning in supporting credit card approval prediction through an interactive web application. The system simplifies applicant evaluation by providing fast and consistent prediction results while maintaining a history of previous predictions. Its modular architecture, responsive interface, and deployment on Render make it suitable for educational demonstrations and future enhancements.

Future improvements may include user authentication, advanced analytics, cloud database integration, model retraining, and enhanced explainable AI features.